

Sample homework: induction

Question: Prove that $1 + 2 + \dots + n = n(n + 1) / 2$ for every integer $n \geq 1$.

Student's submitted proof:

1. Base case: for $n = 1$, both sides equal 1.
2. Assume the formula is true for $k + 1$.
3. Therefore it is true for $k + 1$, which completes the induction.

This submission intentionally contains a mistake for testing.